LIFTING ALGEBRA MAPS TO DIVIDED POWER DGA MAPS OUT OF A TATE RESOLUTION

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Editorial note. The induction step ends 'We omit the details': the omitted content is the mechanical repetition of the displayed degree-1 step at higher degrees (same universal-property application, same acyclicity input).

Allowed standard prerequisites (editorial).

- Universal property of DP polynomial algebras. lemma-divided-power-polynomial-algebra (same chapter, named). Produces the lift A(1) -> B mapping the DP variables to chosen preimages.

Source reference. Original definition 09PJ: sources/stacks-dpa/pinned-original-source.tex, source lines 1300–1300. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 07H5: sources/stacks-dpa/pinned-original-source.tex, source lines 951–951. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let R be a commutative ring. Suppose that (A, d, γ) and (B, d, γ) are as in Definition 09PJ. Let $\overline{\varphi} : H_0(A) \to H_0(B)$ be an R-algebra map. Assume*

(1) *A is a graded divided power polynomial algebra over R.*
(2) *$H_k(B) = 0$ for $k > 0$.*

Then there exists a map $\varphi : A \to B$ of differential graded R-algebras compatible with divided powers that lifts $\overline{\varphi}$.

Proof. The assumption means that A is obtained from R by successively adjoining some set of polynomial generators in degree zero, exterior generators in positive odd degrees, and divided power generators in positive even degrees. So we have a filtration $R \subset A(0) \subset A(1) \subset \ldots$ of A such that $A(m+1)$ is obtained from $A(m)$ by adjoining generators of the appropriate type (which we simply call "divided power generators") in degree $m + 1$. In particular, $A(0) \to H_0(A)$ is a surjection from a (usual) polynomial algebra over R onto $H_0(A)$. Thus we can lift $\overline{\varphi}$ to an R-algebra map $\varphi(0) : A(0) \to B_0$.

Write $A(1) = A(0)\langle T_j : j \in J\rangle$ for some set J of divided power variables T_j of degree 1. Let $f_j \in B_0$ be $f_j = \varphi(0)(\mathrm{d}(T_j))$. Observe that f_j maps to zero in $H_0(B)$ as $\mathrm{d}T_j$ maps to zero in $H_0(A)$. Thus we can find $b_j \in B_1$ with $\mathrm{d}(b_j) = f_j$. By

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the universal property of divided power polynomial algebras from Lemma 07H5, we find a lift $\varphi(1) : A(1) \to B$ of $\varphi(0)$ mapping T_j to f_j.

Having constructed $\varphi(m)$ for some $m \geq 1$ we can construct $\varphi(m+1) : A(m+1) \to B$ in exactly the same manner. We omit the details. $\square$